

Jack Smith

Computer Science · AI, Computer Graphics & Full-Stack Engineering

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PROFILE

Final-year **Computer Science** student at the University of Sheffield with a strong builder's mindset and a focus on **AI, computer graphics and interactive systems**. I design and ship technically demanding software — from full-stack applications for real clients to low-level graphics pipelines and generative-AI tools. My dissertation explores **text-driven 3D human motion generation and editing**. I'm most drawn to problems where AI systems must stay **controllable, usable and expressive** in real-world contexts.

EXPERIENCE

Co-Founder & Full-Stack Developer

2024 — Present

Freelance Web Studio — Sheffield

- Co-founded a small web-development studio delivering modern, responsive websites to local Sheffield businesses with outdated web presences.
- Owned projects end-to-end — client discovery, design, full-stack build and deployment — using React, Tailwind/CSS, GSAP and Cloudflare/Vercel.
- Built B2B sales, client-communication and discovery skills by pitching services, writing proposals and maintaining long-term customer relationships.

Missions Engineer

2023 — 2024

Project Falcon — IMechE Student Team

- Team designed and built a fixed-wing quadcopter to compete in the IMechE UAS Challenge.
- Developed the Flight Ground Control System (FGCS) in Node.js and Python to monitor the drone, record and display telemetry, and receive real-world data live.

SELECTED PROJECTS

HabitFlow — Full-Stack Habit & Goal Platform

2026 · Live

- Full-stack habit and goal tracker with social feeds, AI-powered insights (Google Gemini) and a GitHub-style consistency heatmap.

React TypeScript Express PostgreSQL / Neon Firebase Auth Gemini AI

Neighbourly — Community Help Network · HackSheffield 9 Winner

2026 · Live

- Map-based app matching neighbours needing help with nearby volunteers, with real-time chat. Evolved from our 1st-place (Best Repository) HackSheffield 9 build.

SvelteKit Flask PostgreSQL Google Maps API Auth0

Text-Driven 3D Human Motion Editing Tool · Dissertation

2025 — 2026

- Unity-integrated tool for generating and editing AI-generated 3D human motion from text prompts.
- Enables keyframe-based local regeneration while preserving semantic intent, continuity and physical plausibility.

Unity Generative AI Python 3D Motion

MNIST From-Scratch Classifier & 3D Rasterizer Engine

2023 — 2024

- Built a complete ML pipeline from scratch in Python/NumPy — custom SVM & KNN, PCA and augmentation — reaching 99.8% accuracy on noisy test data.
- Implemented a full software 3D graphics pipeline from scratch (projection, rasterization, z-buffering, .obj loading) to understand how GPUs work at a low level.

Python NumPy / SciPy Computer Vision Computer Graphics

EDUCATION

BSc Computer Science — The University of Sheffield

Modules across AI, Software Engineering & Mathematics

First-Class (on track) · 2026

Consistently strong coursework and exam performance. Dissertation in text-driven 3D human motion generation and editing.

John F. Kennedy School & Sixth Form — Hemel Hempstead

A-Levels & 10 GCSEs (incl. Maths, English, Computer Science)

2016 — 2023

A-Levels: Maths (A*), Computer Science (A), Further Mathematics (B), Physics (B).

SKILLS

LANGUAGES

Python TypeScript JavaScript
Haskell Ruby SQL

AI / ML

PyTorch Generative AI Computer Vision
On-device Inference From-scratch ML

GRAPHICS & 3D

Unity Three.js Shaders Rasterization
Blender

WEB & FULL-STACK

React Next.js SvelteKit Node.js
Flask Rails Tailwind GSAP

DATA & DEVOPS

PostgreSQL Firebase Docker Vercel
Cloudflare Git

HONOURS & AWARDS

- HackSheffield 9 — Best Repository**
1st place · Nov 2024
- Engineering You're Hired**
3rd of cohort · swarm-robotics pipe inspection · 2025
- Dragon's Apprentice + Creativity Prize**
Led team turning £100 → £1,000+ for charity
- Regional Public Speaking**
Runner-up before senior public officials
- SELSA Leadership Academy**
Led 3 interdisciplinary engineering teams
- Snapchat Lens Competition**
1M+ uses · £1,500 prize

LANGUAGES

English — Native
Arabic — Intermediate

BEYOND ENGINEERING

- Bouldering** — outdoor/indoor to f7B+ across the Peak, Lake District, Portland & Fontainebleau.
- Charity hitchhiking** — Sheffield→Porto, Sheffield→Bratislava & Croatia→Morocco; society has raised £1.3M+.
- Judo** — competed nationally at BUCS.
- Guitar** — Grade 8 qualified.

References & portfolio available at www.jacksmith.me